Practice Problems

Problem 4.1

You invest \$100 at time 0, at an annual simple interest rate of 9%.

(a) Find the accumulated value at the end of the fifth year.

(b) How much interest do you earn in the fifth year?

Solution

For simple interest, the accumulation function is

$$a(t) = 1 + it.$$

Here,

$$A(0) = 100, \qquad i = 9\% = 0.09.$$

(a) The accumulated value at the end of the fifth year is

$$FV = 100\big(1 + 0.09(5)\big).$$

Therefore,

$$FV = 145.$$

(b) First compute the accumulated value at the end of the fourth year:

$$FV = 100\big(1 + 0.09(4)\big) = 136.$$

Then the interest earned in the fifth year is

$$I_5 = A(5) - A(4) = 145 - 136 = 9.$$